



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01207010**

FORM TP2012044

MAY/JUNE 2012

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

BIOLOGY

Paper 01 - General Proficiency

75 minutes

06 JUNE 2012 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 75 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Malaria
- (B) Diabetes
- (C) Influenza
- (D) Anaemia

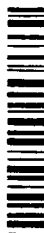
Sample Answer



The best answer to this item is “Anaemia”, so answer space (D) has been shaded.

5. If you want to change your answer, erase it completely and fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to this item later. Your score will be the total number of correct answers.
7. Figures are not necessarily drawn to scale.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.



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1. Which of the following is the correct sequence of feeding relationships in a food chain?

- (A) Herbivore → producer → carnivore → decomposer
- (B) Producer → herbivore → carnivore → top carnivore
- (C) Top carnivore → carnivore → herbivore → producer
- (D) Producer → carnivore → herbivore → scavenger

2. Which of the following are involved in the cycling of nutrients in nature?

- I. Bacteria
- II. Plants
- III. Earthworms
- IV. Humans

- (A) I and II only
- (B) III and IV only
- (C) I, II and III only
- (D) I, II, III and IV

Item 3 refers to the following list.

- I. Number of legs
- II. Body temperature
- III. Shape of body
- IV. Type of body covering

3. The dead remains of a previously unknown animal are found in a forest. Which of the features listed above could be used to classify the animal?

- (A) I and II only
- (B) III and IV only
- (C) I, III and IV only
- (D) II, III and IV only

Item 4 refers to relationships between the following pairs of organisms.

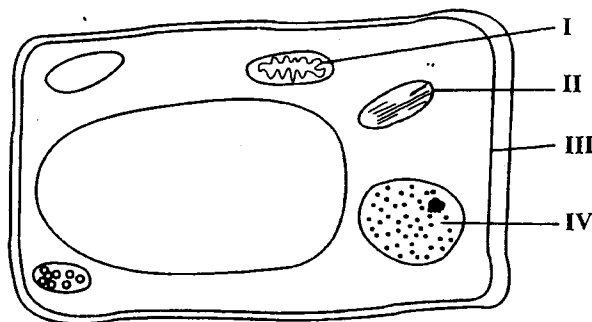
- I. Barnacles on a shark's back
- II. Shark/man
- III. Man/malaria protozoan

4. Which of the following correctly identifies these relationships?

- | | I | II | III |
|-----|---------------|---------------|------------|
| (A) | Commensalism | Predator/prey | Mutualism |
| (B) | Commensalism | Predator/prey | Parasitism |
| (C) | Predator/prey | Commensalism | Mutualism |
| (D) | Predator/prey | Mutualism | Parasitism |

5. Which of the following statements about a food chain is true?
- (A) Energy from the sun is transferred from one organism to subsequent organisms.
 - (B) Each of its members depends on heat energy obtained directly from the sun.
 - (C) Energy changes from one type to another, that is, from heat to light to chemical.
 - (D) The sun traps light energy at the start of the food chain.
6. Which of the following is NOT true about decomposers?
- (A) They release nutrients for recycling.
 - (B) They reproduce faster in low temperatures.
 - (C) They are more abundant in damp places.
 - (D) They prevent the accumulation of dead organic matter.
7. Which of the following organelles is the site of aerobic respiration?
- (A) Nucleus
 - (B) Cytoplasm
 - (C) Chloroplast
 - (D) Mitochondrion

Item 8 refers to the following diagram of a typical plant cell.



8. Which of the labelled parts is NOT found in an animal cell?
- (A) I
 - (B) II
 - (C) III
 - (D) IV
9. Plant cells do not burst when placed in water because
- (A) all the water that enters the cell can be stored in a vacuole
 - (B) plant cells are surrounded by cellulose cell walls
 - (C) ions and other substances in solution move out of the cell as soon as water enters
 - (D) as water enters one cell it immediately passes into an adjacent cell along an osmotic gradient

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10. Which of the following combinations correctly shows a gas that is a raw material and a gas that is a product of photosynthesis?

	Raw Material	Product
(A)	Carbon dioxide	Oxygen
(B)	Carbon dioxide	Nitrogen
(C)	Oxygen	Carbon dioxide
(D)	Oxygen	Nitrogen

11. Which of the following are NOT autotrophs?

- (A) Algae
- (B) Mosses
- (C) Ferns
- (D) Fungi

12. Which of the following occurs as a result of a deficiency of nitrogen in plants?

- (A) Yellowing of the older leaves of the plants
- (B) Stunted chlorotic plants with reduced yields
- (C) Greater susceptibility to diseases and insect attacks
- (D) Dark green overly vigorous foliage leading to reduced yields

Item 13 refers to the table below which shows the observations when a food was tested using different reagents.

Test	Observation
Iodine	Blue-black colour
Sodium hydroxide and copper sulphate (Biuret reagent)	Purple colour
Benedict's or Fehling's solution and heating	No colour change

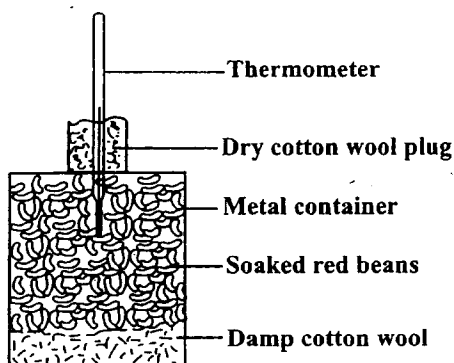
13. The food MOST likely contained

- (A) starch and protein only
- (B) reducing sugar and starch only
- (C) reducing sugar, starch and protein
- (D) protein, fat and non-reducing sugar

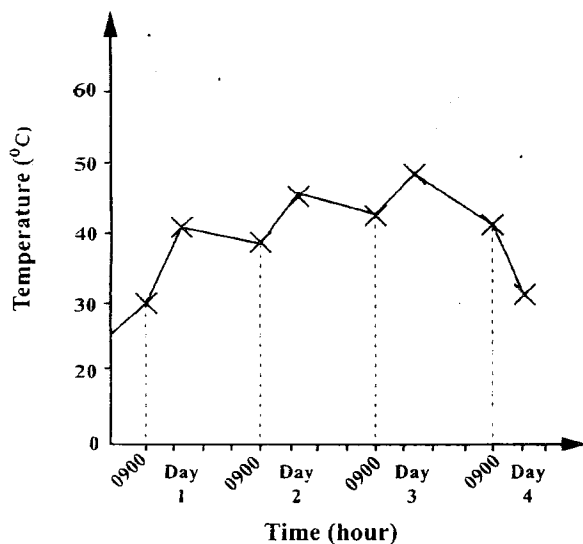
14. Which of the following options is a function of hydrochloric acid in the stomach?

- (A) Emulsifying fat
- (B) Breaking down starch
- (C) Hydrolysing complex sugars
- (D) Providing a medium for pepsin

Item 15 refers to the following apparatus used in an investigation.



The initial temperature of the beans was noted and readings were taken at regular intervals twice daily for four days. The data obtained were used to plot the graph below.



15. The increase in temperature in the vessel is BEST explained by the fact that the seeds are

(A) numerous
(B) taking in water
(C) placed close together
(D) using their food stores

16. Which of the following is NOT a reason for placing green plants in an aquarium?

(A) Providing food for the animals
(B) Providing oxygen for the animals
(C) Removing the chlorine from the water
(D) Removing the animals' waste products

17. Which of the following characteristics is NOT a feature of respiratory surfaces?

(A) Large surface area
(B) Permanently moist
(C) Good supply of blood
(D) Thick membrane

18. The reactants in aerobic respiration are

(A) water and oxygen
(B) glucose and oxygen
(C) water and carbon dioxide
(D) glucose and carbon dioxide

19. An *Amoeba* obtains all the oxygen it needs by diffusion via its cell membrane, while a human needs to have special respiratory surfaces for this purpose. The BEST reason for this difference is that

(A) the *amoeba* does not require much oxygen
(B) oxygen cannot pass through the skin of a human
(C) a human requires a larger volume of oxygen
(D) a human's surface area to volume ratio is too small for diffusion to be effective

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20. Which of the following features is MOST responsible for the efficient uptake of oxygen by red blood cells?

(A) Small size
(B) Softness
(C) Flexibility
(D) Disc shape

21. Which of the following plants does NOT have a modified vegetative storage organ?

(A) Carrot
(B) Ginger
(C) Mango
(D) Potato

22. Mineral salts in plants are transported MAINLY through the

(A) cambium
(B) sieve tubes
(C) parenchyma
(D) xylem vessels

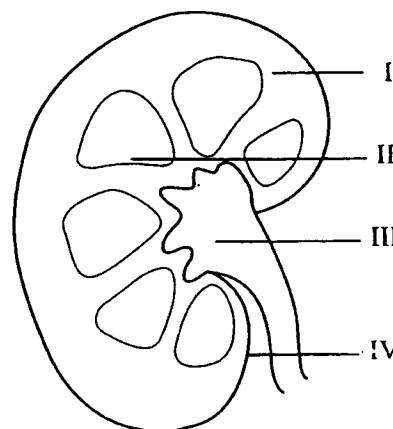
23. Which of the following is NOT a form of excretion in plants?

(A) Carbon dioxide diffusing in, through the stomata
(B) Substances being stored in the bark of a tree
(C) Water vapour diffusing out, through the epidermis
(D) Stored calcium oxalate crystals in leaves during leaf fall

24. Which of the following is NOT an excretory product in humans?

(A) Urea
(B) Water
(C) Undigested food
(D) Sodium chloride

Items 25–26 refer to the following diagram of a mammalian kidney.



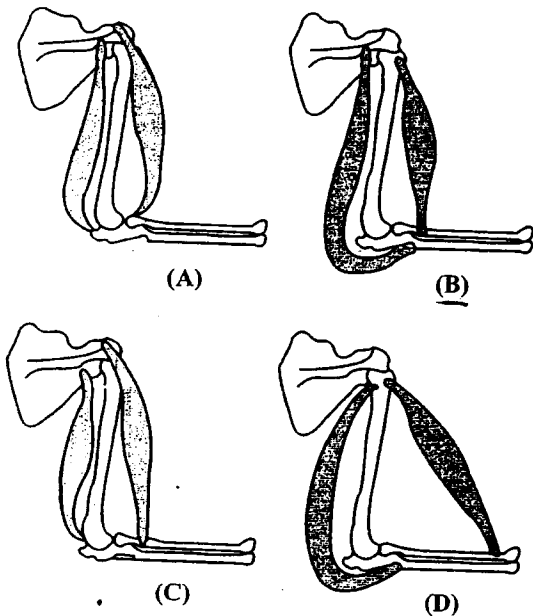
25. In which part of the kidney is the loop of Henlé located?

(A) I
(B) II
(C) III
(D) IV

26. Into which structure is urine emptied?

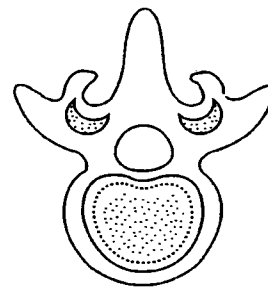
(A) I
(B) II
(C) III
(D) IV

Item 27 refers to the following diagrams.



27. Which of the muscle-bone attachments shown above is accurate?
28. The MAIN function of cartilage at the end of long bones is to
- (A) protect nerves
 - (B) lubricate joints
 - (C) reduce friction
 - (D) increase flexibility
29. Movement is important to the survival of a plant because it ensures that it
- I. can find food
 - II. grows towards light
 - III. grows towards water
 - IV. can respire
- (A) I only
 - (B) I and IV only
 - (C) II and III only
 - (D) I, II, III and IV

Item 30 refers to the following diagram of a vertebra.



30. What type of vertebra is shown above?
- (A) Sacral
 - (B) Lumbar
 - (C) Cervical
 - (D) Thoracic

Item 31 refers to the following sequence of events that occurs when a student reacts to touching a hot object by withdrawing her hand.

Hot object → Pain → Effector → Response

31. The receptor is the
- (A) hot object touched
 - (B) heat from the hot object
 - (C) sensory endings in the skin
 - (D) muscles that contract

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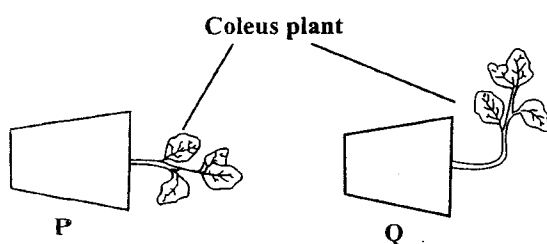
Item 37 refers to the following simple experiment carried out by a student to investigate growth in corn seedlings. After three days the results were as follows.

Plant	1	2	3	4	5	6	7	8	9	10
Change in height (cm)	2.5	2.2	2.1	1.4	2.5	2.5	2.0	2.2	2.8	2.3

37. What is the average change in height, in centimetres, of the corn seedlings during the three days?

- (A) 2.00
- (B) 2.25
- (C) 2.50
- (D) 2.75

Item 38 refers to the following diagram which shows the growth of a plant after a few days in sunlight.



38. The stem at Q grows upward because

- (A) it is attracted to light
- (B) the tip can only grow if it is pointing upward
- (C) auxins accumulated on the upper side of the shoot
- (D) auxins accumulated on the lower side of the shoot

39. Which of the following sequences of events is correct?

- (A) Fertilization → implantation → copulation → gestation
- (B) Fertilization → implantation → gestation → copulation
- (C) Copulation → fertilization → implantation → gestation
- (D) Copulation → implantation → fertilization → gestation

40. The reproduction of *Amoeba* by binary fission is an example of asexual reproduction because

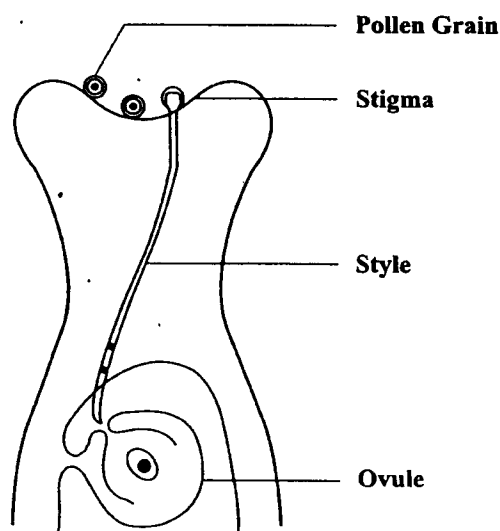
- (A) two offspring are produced by one parent
- (B) the nucleus is not involved in the cell division
- (C) the nucleus divides by meiosis
- (D) the chromosome number is halved

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41. Which of the following involves the process of **asexual** reproduction?

(A) Wind pollination
(B) Insect pollination
(C) Vegetative propagation
(D) Artificial insemination

Item 42 refers to the following diagram.



42. The MOST appropriate heading for this diagram is

(A) fertilization
(B) ovulation
(C) pollination
(D) reproduction

43. Which of the following comparisons of the male and female reproductive systems is **INCORRECT**?

	Male Reproductive System	Female Reproductive System
(A)	Testis	Ovary
(B)	Glands	Uterus
(C)	Sperm	Egg
(D)	Sperm duct	Fallopian tube

44. Contraception is used **PRIMARILY** to

(A) limit the size of families
(B) cause infertility in humans
(C) prevent the spread of sexually transmitted infections
(D) increase the world's population

45. Which of the following statements confirms that a coconut is a fruit?

(A) It can be eaten.
(B) It is dispersed by water.
(C) It has a plumule and a radicle.
(D) It has a three-layered pericarp.

46. As a result of mitosis each daughter cell has

(A) half the number of chromosomes as the parent
(B) the same number of chromosomes as the parent
(C) twice the number of chromosomes as the parent
(D) a variable number of chromosomes

47. In which of the following does meiosis occur?

(A) A developing embryo
(B) The skin of a mammal
(C) The testis of a mammal
(D) The tip of the shoot of a plant

Item 48 refers to the table below which shows the numbers of brown and green grasshoppers found in equal areas of dry and green grass respectively.

	Brown Grasshoppers	Green Grasshoppers
Dry Grass	19	6
Green Grass	8	25

48. Which of the following is the BEST explanation for these observations?

- (A) More food is available in green areas.
- (B) Green grasshoppers are better adapted.
- (C) Camouflaging protects grasshoppers from predators.
- (D) Brown grasshoppers mutate when placed in green areas.

Item 49 refers to the following options which are four patterns of genetic inheritance.

(A) BB black × bb red →

Bb black	Bb black
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(B) HH normal × SS sickle →

HS normal	HS normal
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(C) Nn normal × n-red-green minus →

Nn normal	N-normal
nn red-green minus	n-red-green minus

(D) Rr pink × Rr pink →

RR red	RR red
Rr pink	Rr pink
rr white	rr white

49. Which pattern will produce a 1:2:1 ratio of phenotypes?

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50. Which of the following is NOT involved in natural selection?

- (A) Grafting
- (B) Meiosis
- (C) Variation
- (D) Mutation

51. Which of the following correctly lists examples of pathogenic, nutritional deficiency and physiological diseases?

	Pathogenic	Nutritional Deficiency	Physiological
(A)	Diabetes	AIDS	Anaemia
(B)	AIDS	Diabetes	Anaemia
(C)	Anaemia	Diabetes	AIDS
(D)	AIDS	Anaemia	Diabetes

52. Which of the following trigger the formation of blood clots?

- (A) Thrombocytes
- (B) Erythrocytes
- (C) Lymphocytes
- (D) Phagocytes

53. Through which of the following can natural immunity be obtained?

- I. Blood transfusion
 - II. Breast milk
 - III. The placenta
- (A) I and II only
 - (B) I and III only
 - (C) II and III only
 - (D) I, II and III

54. A vector is defined as an organism that

- (A) bites human
- (B) causes diseases
- (C) lives on other organisms
- (D) transmits disease organisms

Item 55 refers to the following methods of disease treatment/control.

- I. Improved personal hygiene
- II. Isolation of persons with the disease
- III. Good sanitary practices, for example, proper garbage disposal
- IV. Careful food preparation

55. These methods are used in the treatment/control of a disease that is

- (A) inherited
- (B) infectious
- (C) caused by inadequate supply of nutrients
- (D) caused by malfunction of body systems

56. Fish, snails, watergrass and tadpoles living in a freshwater pond may be described as

- (A) a habitat
- (B) a population
- (C) an ecosystem
- (D) a community

57. Which of the following sampling methods are used to estimate the distribution of species of organisms living in soil?

- I. Nets
- II. Quadrats
- III. Bottles
- IV. Jars

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) III and IV only

58. Two soil components that are derived from the parent rock material are

- (A) sand and humus
- (B) sand and mineral salts
- (C) humus and mineral salts
- (D) clay and microorganisms

59. Which of the following resources is renewed through natural cycles?

- (A) Natural gas
- (B) Bauxite
- (C) Forest
- (D) Gold

60. Which of the following practices does NOT help to conserve the environment?

- (A) Overgrazing
- (B) Crop rotation
- (C) Reafforestation
- (D) Using natural fertilizers

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.